

Ph637: 2025: Final Exam: Total marks 40

1. Give a brief reason in one or two sentences for the following statements (3+2+2+3):

- You have applied to Giant Meter-wave Radio Telescope near Pune (Latitude $\phi = 19^\circ$ N) to observe a radio galaxy whose RA= $6^h 40^m$, and Declination $\delta = -75^\circ$. Your proposal is rejected. Why?
- You are observing a main sequence star behind an interstellar cloud, whose opacity is 0.5. You estimate the actual luminosity of the star by its spectral type, and find that its luminosity to be 60% less than what it should be (after correcting for the distance). Why?
- The gyro-frequency of electron is estimated to be 1000 Hz for some value of the magnetic field, but the electron is found to radiate synchrotron radiation at a frequency at 10^9 Hz. Why?
- Main sequence stars obey an approximate relation $L \propto T^8$; why? [Hint: Combine the mass-luminosity relation for main-sequence stars with the fact that central temperature does not vary much across the range of main-sequence stars, to derive this relation].

2. A white-dwarf-supernova with peak luminosity $L = 10^{10} L_\odot$ is observed to have an apparent magnitude of $m = +20$ at this peak. [2+2+2]

- (a) what is its absolute magnitude M ? (Assume the absolute magnitude of the Sun to be +5 for simplicity)
- (b) What is its distance in parsec and light years?
- (c) How long ago did this supernova explode (in yr)?

3. Estimate the typical energy released in a Type Ia supernova, with the following steps [3+3+3]:

- (a) Consider a carbon white dwarf of mass $1.4 M_\odot$, the Chandrasekhar mass. How many ^{12}C nuclei are there?
- (b) Assume that because of nuclear reaction all these carbon nuclei are converted to ^{56}Ni . The mass of each nickel nucleus is 55.94 amu. Estimate the mass loss and the corresponding energy released.
- (d) Show that with a typical radius of 5000 km, this energy released is larger than the gravitational potential energy. This will imply that the supernova will destroy the white dwarf.

4. We can estimate the maximum mass of a star by arguing that radiation pressure can destabilize a massive star, and an approximate upper limit on mass can be obtained by equating the (non-relativistic) kinetic pressure and radiation pressure. [3+3+3]

- (a) Assume that the gravitational potential energy of a star of mass M and radius R (and volume V) is $\sim GM^2/R$. Use the virial theorem to find an expression for the average pressure, in terms of the total mass and average density ρ .
- (b) Next equate the radiation pressure and the kinetic gas pressure and write an expression for the total pressure, that involves only the average density ρ and fundamental constants.
- (c) Obtain the upper limit on the mass of the star (assuming it to be composed of fully ionized hydrogen, or $\mu = 0.5$).

5. For pulsars, we derived in the class two important expressions for the luminosity: (1) assuming that the power is derived from the loss of its rotational energy and (2) assuming the magnetic field to be misaligned with the rotational axis.

(a) Combine these two equations and write $\dot{\Omega} \propto \Omega^3$.

(b) Assume that this relation can be in general written as $\dot{\Omega} \propto \Omega^n$. Pulsar astronomers call n the 'braking index'. Show that $n = \Omega\ddot{\Omega}/\dot{\Omega}^2$.

(c) Recalling that $\Omega = 2\pi/P$, where P is the pulsar period, show that $n = 2 - \frac{P\ddot{P}}{\dot{P}^2}$.

[Observationally, $\ddot{P}$ has been measured for a few pulsars, and it has been found that $1.4 < n < 3$. This means that spin-down of mis-aligned magnetic field model is the complete story—and there are other details one has to think about!]

[3+3]